

Basics 31: Introduction to Waves in Plasmas

V - Plasma waves and Alfvén physics

Phase, dispersion, polarization, eigenmodes, damping, instability, and plasma wave families

Textbook draft, version 1.0. Plasma waves couple particle inertia, pressure, and electromagnetic restoring forces. This chapter develops a common language for waves before specializing to Alfvén physics: phase and group velocity, polarization, dispersion matrices, continua, damping, and instability. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

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1 Learning objectives

After completing this note, the reader should be able to:

- Identify the ingredients required for wave motion.
- Interpret plane-wave phase and wave vector.
- Derive a dispersion relation from a linear system.
- Distinguish phase velocity, group velocity, and polarization.
- Interpret real and imaginary frequency components.
- Classify major electrostatic and electromagnetic plasma-wave families.

2 Prerequisites and reading strategy

- Basics 5-8, 12, 18, 19, and 29.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 What makes a wave?

A wave requires a perturbable state, inertia or delayed response, and a restoring mechanism. A schematic field equation,

$$\partial_t^2 \psi + \mathcal{L}\psi = 0, \quad (3.1)$$

oscillates when the spatial operator $\mathcal{L}$ has positive eigenvalues. In plasmas, restoring forces may arise from charge separation, pressure, magnetic tension, or gyromotion, while several species provide distinct inertias.

4 Plane-wave phase

A local plane wave is written

$$\psi(\mathbf{x}, t) = \text{Re} \left\{ \tilde{\psi} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \right\}. \quad (4.1)$$

Surfaces of constant phase satisfy

$$\mathbf{k} \cdot \mathbf{x} - \omega t = \text{constant}. \quad (4.2)$$

$\mathbf{k}$ points normal to phase fronts and has units of inverse length. The angular frequency ω has units of inverse time. Sign conventions must be recorded because they determine how growth rates and propagation directions are interpreted.

5 Dispersion relations from linear equations

After linearization and Fourier substitution,

$$\partial_t \rightarrow -i\omega, \quad \nabla \rightarrow i\mathbf{k}, \quad (5.1)$$

a coupled PDE system becomes

$$\mathbf{D}(\omega, \mathbf{k}) \tilde{\mathbf{q}} = 0. \quad (5.2)$$

A nontrivial amplitude requires

$$\boxed{\det \mathbf{D}(\omega, \mathbf{k}) = 0}. \quad (5.3)$$

This is the dispersion relation. Multiple roots describe distinct wave branches; the associated null vectors determine polarization and relative phase among field components.

6 Phase and group velocity

For a one-dimensional branch $\omega(k)$,

$$v_{\text{ph}} = \frac{\omega}{k}, \quad v_g = \frac{d\omega}{dk}. \quad (6.1)$$

Phase velocity tracks constant phase. Group velocity gives the leading propagation speed of a narrow wave packet when the medium varies slowly and dissipation is weak. In anisotropic plasma, $\mathbf{v}_g = \nabla_{\mathbf{k}} \omega$ need not be parallel to $\mathbf{k}$.

7 Polarization

Polarization describes the direction and phase relation of vector perturbations. Examples include:

- longitudinal electrostatic oscillations with $\tilde{\mathbf{E}} \parallel \mathbf{k}$;
- transverse electromagnetic waves with $\tilde{\mathbf{E}} \perp \mathbf{k}$;
- circularly polarized waves in magnetized plasma;
- shear-Alfvén waves with transverse velocity and magnetic perturbations.

A scalar dispersion curve alone does not specify the physical mode.

8 Standing waves, eigenmodes, and continua

Boundaries or spatially varying coefficients replace a freely chosen k by a global eigenproblem:

$$\mathcal{A}\xi = \omega^2 \mathcal{M}\xi. \quad (8.1)$$

A finite domain often produces discrete standing modes. In nonuniform ideal MHD, each magnetic surface can carry its own local Alfvén frequency, generating a continuum. Geometry-induced coupling may open gaps in that continuum where discrete global eigenmodes can exist.

9 Complex frequency, damping, and growth

Using $e^{-i\omega t}$, let

$$\omega = \omega_r + i\gamma. \quad (9.1)$$

Then

$$e^{-i\omega t} = e^{-i\omega_r t} e^{\gamma t}. \quad (9.2)$$

Thus $\gamma > 0$ denotes growth and $\gamma < 0$ damping under this convention. Mechanisms include collisions, phase mixing, continuum damping, Landau damping, and energetic-particle drive. A stable oscillation can still decay because of nonideal or kinetic processes.

10 Representative plasma-wave families

At elementary level:

Langmuir waves: electron charge-separation oscillations near ω_{pe} .

Ion-acoustic waves: compressive electrostatic waves involving electron pressure and ion inertia.

Electromagnetic waves: light-like modes modified by plasma response.

Alfvén waves: magnetic-tension waves propagating primarily along the field.

Magnetosonic waves: compressive coupled pressure-magnetic waves.

The appropriate model depends on frequency ordering, propagation angle, collisionality, and scale.

11 Computational laboratory

Solve several analytic dispersion relations and plot $\omega(k)$, phase velocity, and group velocity. Construct a two-component dispersion matrix and recover both eigenvalues and polarization vectors. Form a wave packet numerically and compare measured envelope speed with $d\omega/dk$.

12 Summary

- Plasma waves combine inertia with electrostatic, pressure, magnetic, or gyroscopic restoring forces.
- Dispersion relations determine allowed frequencies; eigenvectors determine polarization.
- Phase and group velocities describe different aspects of propagation.
- Nonuniform plasmas can support continua and discrete global eigenmodes.
- Complex frequency unifies oscillation, damping, and instability under a stated sign convention.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
k	wave vector	Spatial phase gradient.
ω	complex angular frequency	Oscillation and growth/damping rate.
v_{ph}	phase velocity	ω/k in one dimension.
v_g	group velocity	$d\omega/dk$.
D	dispersion matrix	Fourier-transformed linear operator.
γ	growth rate	$\text{Im } \omega$ for $e^{-i\omega t}$.

13 Exercises

1. Derive the constant-phase propagation velocity of a plane wave.
2. Obtain the wave-equation dispersion relation by Fourier substitution.
3. Explain why $\det D = 0$ is required for a nonzero amplitude.
4. For a given dispersive branch, calculate phase and group velocities.
5. Translate the growth-rate sign if the time convention is changed to $e^{+i\omega t}$.

14 References

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